

VisionAssist Smart Glasses System

Comprehensive Software Stack & Technical Architecture Report

1. Executive Overview

VisionAssist is an AI-powered smart glasses software platform engineered to empower visually impaired and blind users with real-time independence. The system integrates cutting-edge deep learning, computer vision, optical character recognition (OCR), multi-lingual voice feedback, and GPS navigation. This document details all software components, frameworks, AI models, and APIs used in the architecture.

2. Software Stack Summary Table

Category	Software / Library	Version / Provider	Primary Role & Purpose
Deep Learning AI	PyTorch	2.11.0	Custom Currency CNN neural network trained on dataset for high-accuracy note classification.
Generative AI	Google Gemini 2.5 Flash	OpenRouter API	Multimodal AI for holistic scene description, voice Q&A; companion, and currency validation.
Object Detection	YOLOv8 (Ultralytics)	YOLOv8 Nano	Real-time identification of pedestrians, cars, buses, obstacles, traffic lights, and zebra crossings.
Computer Vision	OpenCV & EasyOCR	4.11.0 / EasyOCR	Text recognition from signs/labels, traffic light color spectrum (HSV) detection, and color extraction.
Backend Server	FastAPI & Uvicorn	Python 3.11	High-speed asynchronous microservice engine processing camera frame payloads.
Cloud & Database	Supabase Edge Functions	PostgreSQL / Deno	Serverless cloud proxy, user state management, and configuration store.
Frontend Framework	React 18 & TypeScript	Vite Build Tool	Ultra-responsive user interface, interactive dashboard, HUD visualizers, and state engine.
Maps & GPS	Leaflet & OpenStreetMap	OpenSource GIS	Interactive navigation map panel, geocoding destination lookup, and turn-by-turn routing.
Speech & Voice	Web Speech API & pyttsx3	HTML5 / Offline TTS	Hands-free voice command parsing and spoken voice notifications in multiple languages.

3. Detailed Software Breakdown & Architecture

A. PyTorch Deep Learning Currency Engine

- **Role:** High-speed local neural network for Indian currency note classification.
- **Dataset:** Trained on custom dataset containing █10, █20, █50, █100, █200, █500, and █2000 notes.
- **Architecture:** 4-stage Convolutional Neural Network (CNN) with Batch Normalization, Max Pooling, and Dropout regularization.

- **Performance:** Delivers predictions in under 50ms without requiring internet connection.

B. Google Gemini 2.5 Flash Multimodal AI

- **Role:** Primary cloud vision AI partner.
- **Features:** Provides human-like holistic scene descriptions (e.g. 'You are in a hallway...'), answers open-ended user questions via natural voice Q&A, and translates camera readings.
- **Integration:** Connects via OpenRouter REST endpoint with multi-lingual prompt conditioning.

C. Ultralytics YOLOv8 & OpenCV Engine

- **Role:** Real-time safety hazard & object detector.
- **Capabilities:** Scans video streams at up to 30 FPS for nearby pedestrians, moving vehicles, traffic lights, and zebra crossings.
- **HSV Pipeline:** Processes hue, saturation, and brightness values of traffic lights to confirm RED, YELLOW, or GREEN states.

D. Multi-Lingual Speech & Voice Engine

- **Supported Languages:** English, Tamil (■■■■■■), Hindi (■■■■■■), Telugu (■■■■■■), Kannada (■■■■■■).
- **Text-To-Speech (TTS):** Native browser Web Speech Synthesis for crystal-clear audio announcements.
- **Voice Commands:** Enables hands-free navigation, feature triggers, and emergency SOS alerts.

E. Leaflet GPS Navigation & Geocoding

- **Role:** Turn-by-turn walking guidance.
- **Functionality:** Calculates distance, estimated time of arrival (ETA), current road name, and destination door number matching via OCR.

4. Reliability & Privacy Design

The software architecture enforces a **hybrid offline/online design**. Critical safety features (obstacle warning, currency identification via PyTorch, OCR, traffic light sensing) run locally on-device for ultra-fast latency and continuous operation even in offline areas. Cloud AI models are used to enhance conversational detail when connected.